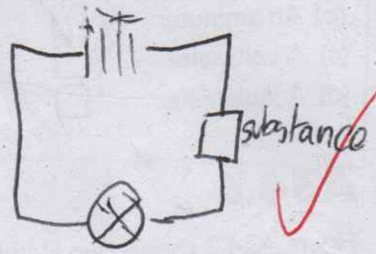
34 Current electricity

A34.1

What is an electric current? The flow of electrons through a circuit

A34.2

(a) Draw a circuit diagram you could use to identify if a substance is a conductor of electricity.



(b) How would you know if a substance is a good conductor of electricity or a poor conductor?

If the light goes on

(c) Underline which of the following conduct electricity:

euro coin	**plastic pen**	**metal paperclip**
brass key	**graphite in a pencil**	**water**

(d) What term is given to a substance that does not conduct electricity? insulator

(e) Give a use for such a substance. protection on electrical wiring

(f) Name the circuit symbols shown in Table A34.1.

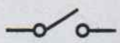
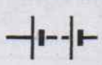
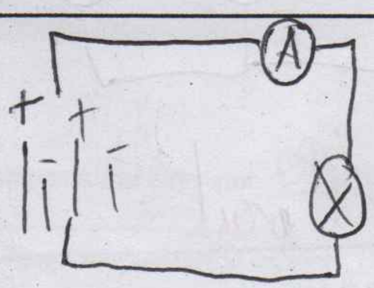
Symbol	Name
	Light emitting diode
	Resistor
	Switch
	Ohmmeter
	Battery

Table A34.1

A34.3

In the box provided draw a circuit diagram of a bulb in a circuit with a battery and an appropriate instrument used to measure the current flowing in the circuit.



A34.4

The circuit symbol in Figure A34.1 is the symbol for:

(a) A fixed resistor ☐
(b) A bulb ☐
(c) A light-emitting diode ☑
(d) An ohmmeter ☐

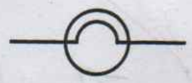


Figure A34.1

A34.5

The instrument used to measure current is called:

(a) An ohmmeter ☐
(b) An ammeter ☑
(c) A voltmeter ☐
(d) A joulemeter ☐

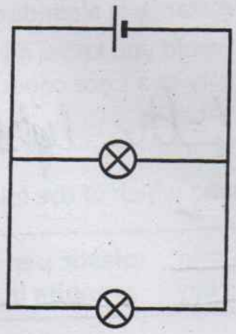


Figure A34.2

A34.6

Figure A34.2 shows two lit bulbs. The bulbs are:

(a) Connected in series ☐
(b) Connected in parallel ☑
(c) Connected one in series and one in parallel ☐
(d) Not connected ☐

A34.7

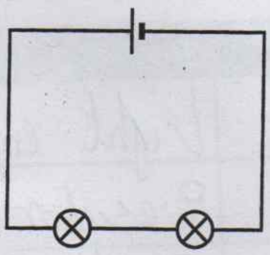


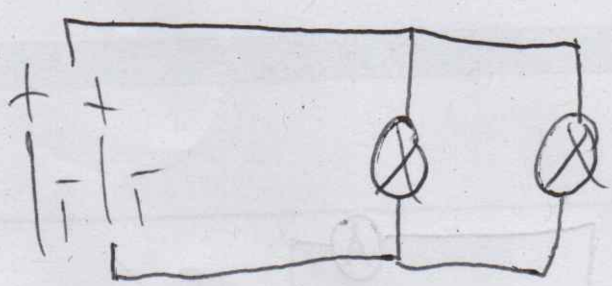
Figure A34.3

(a) What term is used to describe the arrangement of bulbs in Figure A34.3?

Series

(b) If one bulb blows, the other bulb stops lighting. Explain why this is the case. Because current can no longer pass through the blown bulb

(c) Draw an alternative arrangement of the circuit so that if one bulb blows the other will remain lighting.



Parallel Circuit

(d) Give an example of where this lighting arrangement is used. Headlights

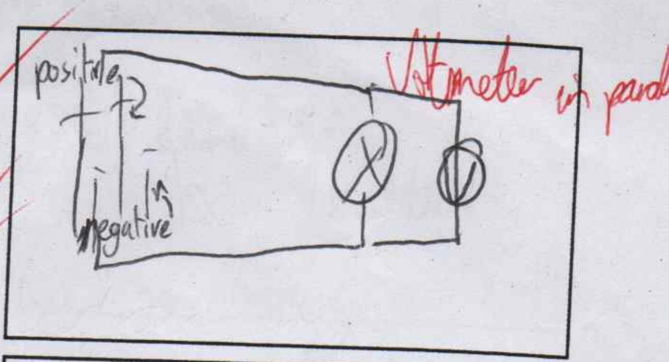
I didn't see this question

A34.8

(a) In the box provided draw a circuit diagram of a bulb and a battery so the bulb lights up.

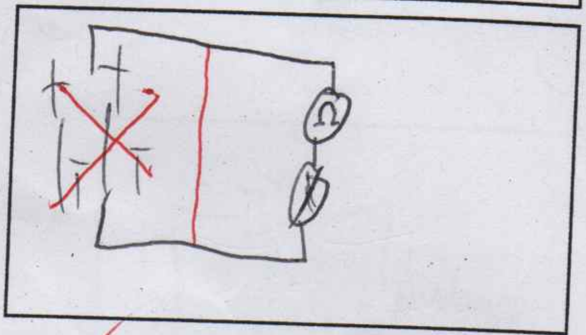
(b) Draw a voltmeter connected correctly to measure the potential difference across the bulb.

(c) In the diagram indicate the positive side and negative side of the battery symbol.



A34.9

(a) Draw a circuit diagram of the apparatus used to measure the resistance of a light bulb.



(b) What are the units of resistance?

Ohm

A34.10

In Figure A34.4, the circuit symbol for a battery is:

(a) Symbol 1 ☐

(b) Symbol 2 ☑

(c) Symbol 3 ☐

(d) Symbol 4 ☐

A34.11

The device used for measuring potential difference = voltage is known as:

(a) A variable resistor ☐

(b) An ammeter ☐

(c) A voltmeter ☑

(d) An ohmmeter ☐

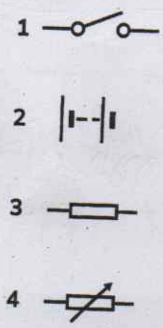


Figure A34.4

A34.12

Name the instruments represented by the symbols shown in Table A34.2.

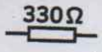
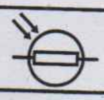
Symbol	Name
330Ω 	~~Resistor~~ Fixed resistor
	Variable resistor
	Light dependent resistor Street lamps
	Thermistor Heat

Table A34.2

A34.13

(a) Name the instrument used to measure the resistance of a resistor. Ohmmeter

(b) What is the unit of resistance? Ohm

(c) For a constant voltage, if the resistance of a circuit increases, the current decreases.